

DM545/DM871 – Linear and integer programming

Exercise Sheet 4, Autumn 2026

Solution:

Included.

Exercises with the symbol $+$ are to be done at home before the class. Exercises with the symbol $*$ will be tackled in class and should be at least read at home. The remaining exercises are left for self training after the exercise class.

Exercise 1 $^+$

Using the bounding method we showed in class that

$$\text{the dual of } \max\{c^T x \mid Ax = b, x \geq 0\} \text{ is } \min\{y^T b \mid y^T A \geq c^T\}.$$

Using the same method show that

$$\text{the dual of } \max\{c^T x \mid Ax \leq b, x \geq 0\} \text{ is } \min\{y^T b \mid y^T A \geq c^T, y \geq 0\}.$$

Solution:

Everything is meant in vector terms even if not in bold. Given $\max\{c^T x \mid Ax \leq b, x \geq 0\}$ we search for multipliers $y \geq 0$ such that $y^T Ax \leq y^T b$ (since we have $\leq$ inequalities, it is important that the multipliers are restricted to be non negative since we want to preserve the direction of the inequality). To ensure that we find an upper bound and hence have $c^T x \leq y^T Ax$, we impose $y^T A \geq c^T$ (since $x \geq 0$). Hence, the best upper bound will be given by solving $\min\{y^T b \mid y^T A \geq c^T, y \geq 0\}$ (recalling from linear algebra that $(AB)^T = B^T A^T$, we can rewrite: $\min\{b^T y \mid A^T y \geq c, y \geq 0\}$, which is the form we would obtain using the recipe method).

Exercise 2 $*$

Consider the following LP problem:

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ & 2x_1 + 3x_2 \leq 30 \\ & x_1 + 2x_2 \geq 10 \\ & x_1 - x_2 \leq 1 \\ & x_2 - x_1 \leq 1 \\ & x_1 \geq 0 \end{aligned}$$

- Write the dual problem
- Using the optimality conditions derived from the theory of duality, and without using the simplex method, find the optimal solution of the dual knowing that the optimal solution of the primal is $(27/5, 32/5)$.

Solution:

The dual is:

$$\begin{aligned} \min \quad & 30y_1 + 10y_2 + y_3 + y_4 \\ & 2y_1 + y_2 + y_3 - y_4 \geq 2 \\ & 3y_1 + 2y_2 - y_3 + y_4 = 3 \\ & y_1, y_3, y_4 \geq 0 \\ & y_2 \leq 0 \end{aligned}$$

We use the complementary slackness theorem.

$$\begin{cases} 2y_1 + y_2 + y_3 - y_4 = 2 \\ 3y_1 + 2y_2 - y_3 + y_4 = 3 \\ y_2 = 0 \\ y_3 = 0 \end{cases}$$

The first because the corresponding variable of the primal is > 0 , the second for the same reason or however because it is already tight by definition, the third and fourth equations are a consequence of the fact that substituting the value of the primal variables in the primal problem, the second and third constraints are binding. What we obtain is a linear system of four equations in four variables that we can solve to find the value of the variables of the dual problem.

Exercise 3

Consider the problem

$$\begin{array}{ll} \text{maximize} & 5x_1 + 4x_2 + 3x_3 \\ \text{subject to} & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

Without applying the simplex method, how can you tell whether the solution $(2, 0, 1)$ is an optimal solution? Is it? [Hint: consider consequences of Complementary slackness theorem.]

Solution:

The primal problem:

$$\begin{array}{ll} \max & 5x_1 + 4x_2 + 3x_3 \\ \text{s.t} & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

The dual problem:

$$\begin{array}{ll} \min & 5y_1 + 11y_2 + 8y_3 \\ \text{s.t} & 2y_1 + 4y_2 + 3y_3 \geq 5 \\ & 3y_1 + y_2 + 4y_3 \geq 4 \\ & y_1 + 2y_2 + 2y_3 \geq 3 \\ & y_1, y_2, y_3 \geq 0 \end{array}$$

If the Complementary slackness theorem holds and the objective value for the primal and dual is the same, then the solution $(2, 0, 1)$ is optimal. We check that it does:

Since $x_1, x_3 > 0$ we have the two constraints:

$$\begin{aligned} 2y_1 + 4y_2 + 3y_3 &= 5 \\ y_1 + 2y_2 + 2y_3 &= 3 \end{aligned}$$

Now we can test which constraints are binding in the primal problem.

We can see that the second constraint is non-binding, then $y_2 = 0$ and we have the following system

$$2y_1 + 4y_2 + 3y_3 = 5$$

of equations: $y_1 + 2y_2 + 2y_3 = 3$

$$y_2 = 0$$

Then we have that:

$$2y_1 + 3y_3 = 5$$

$$y_1 + 2y_3 = 3$$

Solved as before will give $y_1 = 1, y_3 = 1$.

Then the solution is:

$$y_1^* = 1, y_2^* = 0, y_3^* = 1$$

We need to check if it is feasible. We have already checked the conditions for the primal problem and we have already checked the first and third constraint in the dual, so we need to check the second constraint of the dual. It needs to be non-binding since $x_2 = 0$:

$$\begin{aligned} & 3y_1^* + y_2^* + 4y_3^* > 4 \\ \Rightarrow & 3 \cdot 1 + 0 + 4 \cdot 1 > 4 \\ \Rightarrow & 7 > 4 \end{aligned}$$

So it's non-binding.

We check that the objective values are the same:

For the primal: $5 \cdot 2 + 4 \cdot 0 + 3 \cdot 1 = 13$

For the dual: $5 \cdot 1 + 11 \cdot 0 + 8 \cdot 1 = 13$

Then the solution is optimal.

Exercise 4

Consider the following LP:

$$\begin{aligned} \min & 3x_1 + 2x_2 - 4x_3 \\ & 2x_1 + x_2 + x_3 \geq 3 \\ & x_1 + x_2 + 2x_3 \leq 5 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Find the optimal solution knowing that the solution of the dual problem is $(u_1, u_2) = (10/3, 11/3)$.

Solution:

Rewrite:

$$\begin{aligned} \max & -3x_1 - 2x_2 + 4x_3 \\ \text{s.t.} & -2x_1 - x_2 - x_3 \leq -3 \\ & x_1 + x_2 + 2x_3 \leq 5 \\ & x_i \geq 0 \quad \forall i \in \{1, 2, 3\} \end{aligned}$$

The dual problem:

$$\begin{aligned} \min & -3u_1 + 5u_2 \\ \text{s.t.} & -2u_1 + u_2 \geq -3 \\ & -u_1 + u_2 \geq -2 \\ & -u_1 + 2u_2 \geq 4 \\ & u_1, u_2 \geq 0 \end{aligned}$$

Now we want to find a solution to the primal. Since $u_1, u_2 > 0$ then we have the two constraints:

$$\begin{aligned} -2x_1 - x_2 - x_3 &= -3 \\ x_1 + x_2 + 2x_3 &= 5 \end{aligned}$$

We check which constraints are binding in the dual:

We see that the second constraint is non-binding, hence $x_2 = 0$. And we have the following system of equations:

$$\begin{aligned} -2x_1 - x_2 - x_3 &= -3 \\ x_1 + x_2 + 2x_3 &= 5 \\ x_2 &= 0 \end{aligned}$$

Hence we have:

$$\begin{aligned} -2x_1 - x_3 &= -3 \\ x_1 + 2x_3 &= 5 \Rightarrow x_1 = 5 - 2x_3 \end{aligned}$$

Then we can insert x_1 in the first equation:

$$\Rightarrow -2(5 - 2x_3) - x_3 = -3 \Rightarrow -10 + 3x_3 = -3 \Rightarrow 3x_3 = 7 \Rightarrow x_3 = 7/3$$

And we can find x_1 :

$$x_1 = 5 - 2x_3 = 5 - 2 * 7/3 = 1/3$$

Then the solution of the primal is: $x_1 = 1/3, x_2 = 0, x_3 = 7/3$

Exercise 5*

Write the dual of the following problem

$$\begin{aligned}
 (P) \quad & \max \sum_{j \in J} \sum_{i \in I} r_j x_{ij} \\
 & \sum_{j \in J} x_{ij} \leq b_i & \forall i \in I \\
 & \sum_{i \in I} x_{ij} \leq d_j & \forall j \in J \\
 & \sum_{i \in I} p_i x_{ij} = p_j \sum_{i \in I} x_{ij} & \forall j \in J \\
 & x_{ij} \geq 0 & \forall i \in I, j \in J
 \end{aligned}$$

Solution:

There are three different sets of constraints. We introduce the dual variables $\alpha_i \geq 0$, for $i \in I$, for the first set; the dual variables $\beta_j \geq 0$, for $j \in J$, for the second set; and the dual variables $\gamma_j \in \mathbb{R}$, for $j \in J$ for the third set.

We then write the A matrix for the example in the picture, augmented with the b vector:

$$\begin{array}{cccccccc}
 & x_{G1} & x_{G2} & x_{G3} & x_{G4} & x_{T1} & x_{T2} & x_{T3} & x_{T4} & \\
 \left[\begin{array}{cccccccc}
 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & b_1 \\
 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & b_2 \\
 1 & & & & 1 & & & & d_1 \\
 & 1 & & & & & 1 & & d_2 \\
 & & 1 & & & & & 1 & d_3 \\
 & & & 1 & & & & & 1 & d_4 \\
 p_G - p_1 & & & & & p_T - p_1 & & & & \\
 & p_G - p_2 & & & & & p_T - p_2 & & & \\
 & & p_G - p_3 & & & & & p_T - p_3 & & \\
 & & & p_G - p_4 & & & & & p_T - p_4 &
 \end{array} \right]
 \end{array}$$

and finally the dual from the columns of the A matrix in general terms:

$$\begin{aligned}
 (D) \quad & \min \sum_{i \in I} \alpha_i b_i + \sum_{j \in J} \beta_j d_j \\
 & \alpha_i + \beta_j + (p_i - p_j) \gamma_j \geq r_j & \forall i \in I, j \in J \\
 & \alpha_i \geq 0 & \forall i \in I \\
 & \beta_j \geq 0 & \forall j \in J \\
 & \gamma_j \in \mathbb{R} & \forall j \in J
 \end{aligned}$$

Exercise 6* Level Terrain for a New Highway

The Highway Department is planning a new 10-km highway on uneven terrain as shown by the profile in Figure 1. The width of the construction terrain is approximately 50 meters and the differences in height measured in meters can be deduced from the y -axis in the figure. To simplify the situation, the terrain profile can be replaced by a step function as shown in the figure. Using heavy machinery, earth removed from high terrain is pulled with effort to fill low areas. There are also two burrow pits, I and II, located at the ends of the 10-km stretch from which additional earth can be pulled, if needed. Pit I is located before km 0 and has a capacity of 15 000 cubic meters, and pit II is located after km 10 and has a capacity of 11 000 cubic meters. The costs of removing earth from pits I and II are, respectively, 15 kr and 19 kr per cubic meter.

The transportation cost per cubic meter per kilometer is 1.5 kr, and the cost of using heavy machinery to load pulling trucks is 2 kr per cubic meter. This means that, for example, three cubic meters extracted from pit I and transported for 1 km will cost a total of $15 \cdot 3 + 1.5 \cdot 3 \cdot 1 + 2 \cdot 3 = 96$ kr and three cubic meters pulled 1 km away from a hill to a fill area will cost $1.5 \cdot 3 \cdot 1 + 2 \cdot 3 = 55.5$ kr.

Write a linear programming model for finding the minimum cost plan for leveling the 10-km stretch.

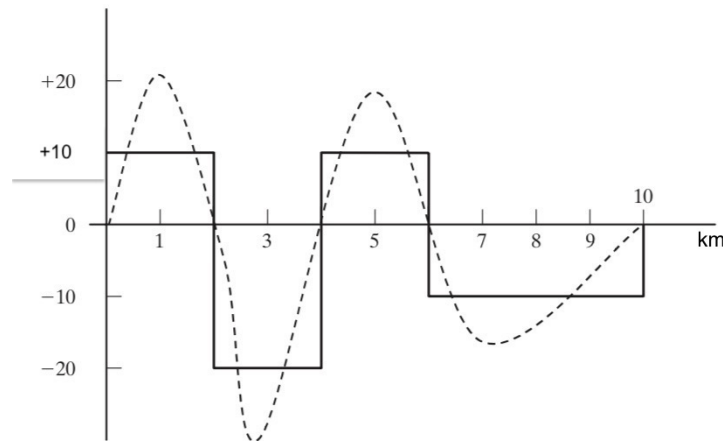


Figure 1:

Solution:

Let $i \in \{1..10\}$ denote the vertical parallelepiped with basis the segment of coordinates $i-1, i$ of length 1000 m. Let 0 indicate the pit I and 11 the pit II. Let a_i be the volume of the parallelepiped i . Let $I^+ \subset \{0..11\}$ be the set of parallelepiped that represent hill areas and $I^- \subset \{0..11\}$ be the set of those representing a fill area. The values of a_i can be calculated by calculating the volumes using the measures from the figure after proper transformation from kilometers to meters and by using the information about the width of 50 meters given in the problem. Let also $a_0 = 15000$ and $a_{11} = 11000$. Let d_{ij} be the distance between the centers of the parallelepiped i and j . The values of d_{ij} are expressed in meters and can be deduced from the figure.

Let $x_{ij} \geq 0, i, j = 1..10$ be the earth moved from parallelepiped i to position j .

$$\min \sum_{i=1}^{10} \sum_{j=1}^{10} (2 + 1.5 \cdot d_{ij}) x_{ij} + \sum_{j=1}^{10} (17 + 1.5 d_{0j}) x_{0j} + \sum_{j=1}^{10} (21 + 1.5 d_{11,j}) x_{11,j}$$

$$\sum_{i=0}^{11} x_{ij} = a_j \quad \forall j \in I^-$$

$$\sum_{j=1}^{10} x_{ij} \leq a_i \quad \forall i \in I^+$$

$$x_{ij} \geq 0 \quad \forall i, j = 1..10$$

The objective function measures the cost of moving earth around. The first component calculates the cost between the areas and the last two the cost of moving earth from the pit I and II, respectively. The first set of constraints ensure that the earth moved to the areas of low terrain is enough to fill the holes. The second set of constraints ensure that the earth moved away from the high areas does not make them become low areas. Finally, all earth movements are non negative.

Exercise 7*

Consider the following problem:

$$\begin{array}{ll}\text{maximize} & z = x_1 - x_2 \\ \text{subject to} & x_1 + x_2 \leq 2 \\ & 2x_1 + 2x_2 \geq 2 \\ & x_1, x_2 \geq 0\end{array}$$

In the ordinary simplex method this problem does not have an initial feasible basis. Hence, the method has to be enhanced by a preliminary phase to attain a feasible basis. Traditionally we talk about a *phase I–phase II* simplex method. In phase I an initial feasible solution is sought and in phase II the ordinary simplex is started from the initial feasible solution found.

Phase I can be done by solving an auxiliary LP problem defined by introducing auxiliary variables and minimizing them in the objective. The solution of the auxiliary LP problem gives an initial feasible basis or a proof of infeasibility.

Apply the phase I–phase II simplex method (that is, phase I is carried out with the auxiliary problem) to the problem above.

Solution:

$$\begin{array}{rcll}\max & x_1 & - & x_2 & = & z \\ & x_1 & + & x_2 & \leq & 2 \\ & 2x_1 & + & 2x_2 & \geq & 2 \\ & & & & & x_1, x_2 \geq 0\end{array}$$

We put in equational standard form by introducing a slack variable $s_1 \geq 0$ and a surplus variable $s_2 \geq 0$:

$$\begin{array}{rcll}\max & x_1 & - & x_2 & = & z \\ & x_1 & + & x_2 & + & s_1 & = & 2 \\ & 2x_1 & + & 2x_2 & - & s_2 & = & 2 \\ & & & & & & & x_1, x_2, s_1, s_2 \geq 0\end{array}$$

This form is not canonical and therefore the first tableau does not have a feasible starting solution.

Auxiliary Problem Approach

We proceed by

- Phase I solving an auxiliary/augmented problem
- Phase II continuing with ordinary simplex

Phase I We introduce an auxiliary variable $a_1 \geq 0$ in the constraint that makes the infeasibility, thus yielding a canonical form:

$$\begin{array}{rcll}\max & x_1 & - & x_2 & = & z \\ & x_1 & + & x_2 & + & s_1 & = & 2 \\ & 2x_1 & + & 2x_2 & - & s_2 & + & a_1 & = & 2 \\ & & & & & & & & x_1, x_2, s_1, s_2, a_1 \geq 0\end{array}$$

Now we have a canonical form

x1	x2	s1	s2	a1	-z	b
1	1	1	0	0	0	2
2	2	0	-1	1	0	2
1	-1	0	0	0	1	0

This problem will have the same solution as the original one only when $a_1 = 0$. We can then solve

- an *augmented problem* by introducing the following objective function $\max w = x_1 - x_2 - Ma_1$, where M is a large enough constant or
- an *auxiliary problem* $\min w = a_1 = -\max(-a_1)$.

Let's take the auxiliary problem, if $w^* > 0$ then we will conclude that the feasibility region of the original problem is empty. Otherwise, if $w^* = 0$, then this implies that $a_1 = 0$ and we found a feasible solution. Let's proceed by setting up the tableau of the auxiliary problem

x1	x2	s1	s2	a1	-z	-w	b
1	1	1	0	0	0	0	2
2	2	0	-1	1	0	0	2
1	-1	0	0	0	1	0	0
0	0	0	0	-1	0	1	0

This is not in canonical form but it is easy to bring it to canonical form: just add the second row to the last one.

x1	x2	s1	s2	a1	-z	-w	b
1	1	1	0	0	0	0	2
2	2	0	-1	1	0	0	2
1	-1	0	0	0	1	0	0
2	2	0	-1	0	0	1	2

The variables s_1, a_1 give us a feasible basis now. It is not optimal. We proceed with the pivot operations. In this case it is worth noting that in the ratio rule, we do not consider the third row since that row corresponds to the original objective function and not to a constraint.

We make x_1 enter the basis and consequently a_1 goes out. The pivot is 2 and the new tableau:

	x1	x2	s1	s2	a1	-z	-w	b
R1' = R1 - R2'	0	0	1	1/2	-1/2	0	0	1
R2' = R2/2	1	1	0	-1/2	1/2	0	0	1
R3' = R3 - R2'	0	-2	0	1/2	-1/2	1	0	-1
R4' = R4 - R2	0	0	0	0	-1	0	1	0

The tableau is optimal. One non basic variable has reduced cost null, which indicates that there are infinite solutions, but this is not relevant now. The relevant thing is that $w^* = 0$ hence the minimum of the auxiliary problem is 0 and hence there is a feasible solution for $a_1 = 0$. This concludes the Phase I of the algorithm since a feasible solution for the auxiliary problem is feasible also for the original problem.

Phase II We throw away the last row and the second last column from the tableau since we do not need them anymore.

	x1	x2	s1	s2	a1	-z	b
	0	0	1	1/2	-1/2	0	1
	1	1	0	-1/2	1/2	0	1
	0	-2	0	1/2	-1/2	1	-1

The tableau is not optimal. The basic solution corresponding to this tableau is feasible but not optimal. We bring s_2 in the basis and make s_1 leave. The new tableau is:

	x1	x2	s1	s2	a1	-z	b
R1' = 2*R1	0	0	2	1	-1	0	2
R2' = R2 + R1	1	1	0	0	1/2	0	2
R3' = R3 - R1	0	-2	-1	0	0	1	-2

The tableau is now optimal. The optimal solution is $x = (2, 0)$ and $z^* = 2$.

We can visualize the problem using the LP Grapher tool linked from the course webpage:

Enter the linear programming problem here:

☒ Maximize $z = x - y$ subject to the constraints:

☐ Minimize

☐ Show only the region defined by the following constraints:

$x + y \leq 2$
 $2x + 2y \geq 2$

LP Examples Graphing Examples Solve

Rounding: 4 decimal places Fraction Mode ☐

Erase Everything

The solution will appear below.

Vertex	Lines through vertex	Value of objective
• (0, 2)	$x + y = 2$ $x = 0$	-2
• (2, 0)	$x + y = 2$ $y = 0$	2 Maximum
• (0, 1)	$2x + 2y = 2$ $x = 0$	-1
• (1, 0)	$2x + 2y = 2$ $y = 0$	1

